

EPART_L2

1. Analysis of datasets

- Summary statistics (only true features; label column excluded)
 - Data before cleansing

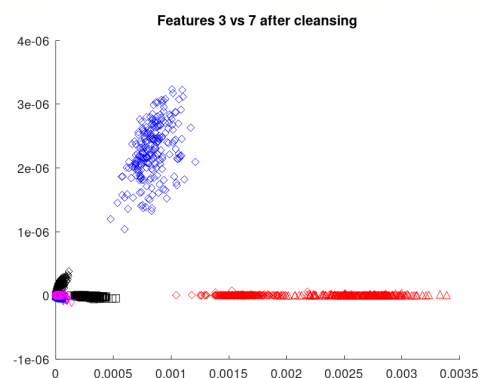
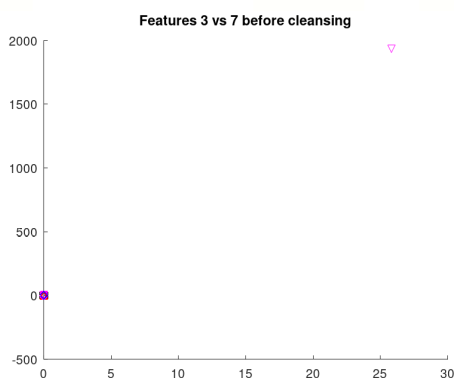
Statistic	Feature 1	Feature 2	Feature 3	Feature 4	Feature 5	Feature 6	Feature 7
Mean	1.87E-01	1.48E-02	2.10E-01	2.09E-01	7.97E+01	1.06E+00	9.08E-03
Median	1.83E-01	1.48E-04	1.74E-04	2.00E-06	-8.94E-11	1.36E-10	-1.84E-14

Setup. We first inspect the training set, remove two extreme outliers (one max-type and one min-type sample using the course's hint procedure), and select two features for the rest of the experiments.

- Data after cleansing

Statistic	Feature 1	Feature 2	Feature 3	Feature 4	Feature 5	Feature 6	Feature 7
Mean	1.84E-01	6.83E-04	9.49E-04	1.47E-05	-6.98E-10	2.97E-07	-9.47E-11
Median	1.83E-01	1.48E-04	1.74E-04	2.00E-06	-9.08E-11	1.36E-10	-1.87E-14

Observation. After removing two clear outliers, feature means moved closer to the medians while medians barely changed. This indicates reduced skewness due to extreme samples. Such cleansing stabilizes Gaussian parameter estimates and should improve the robustness of downstream classifiers.



2. Chosen features: Feature 3 and Feature 7.

3. Baseline Bayes classification with equal

priors (Point 2)

- a. **Priors.** 8 classes, each prior = 0.125.

Error rates (test set, using F3 and F7):

Classifier	Error
Independent Gaussian	0.1623
Multivariate Gaussian	0.318
Parzen window ($h_1 = 0.001$)	0.318

Comment: With the current two-feature projection, the independent Gaussian model performs best. The multivariate Gaussian underperforms—likely because in two dimensions the empirical covariance estimates are sensitive to class shape and residual outliers. The Parzen result at $h_1=0.001$ on the full test set aligns with the multivariate baseline here.

4. Impact of reducing the TRAIN set equally across classes (Point 3)

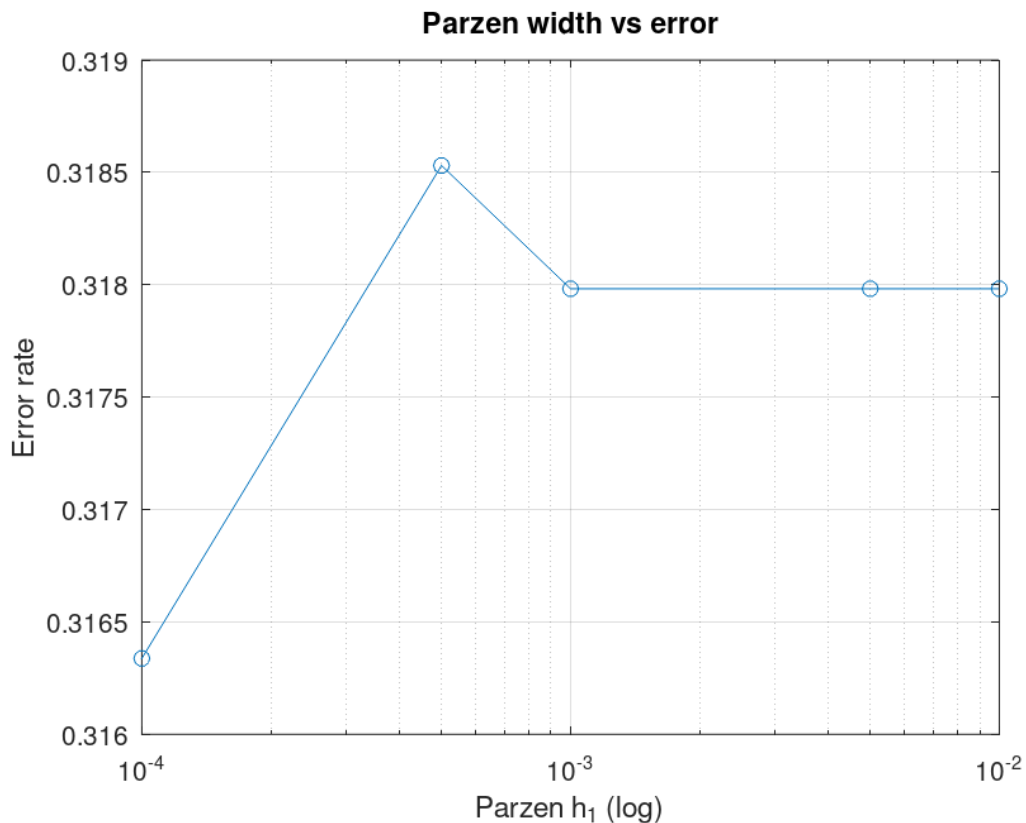
- a. **Settings.** Reduction ratios per class: parts = [0.1 0.25 0.5]; . Repetitions: 5. We report **mean $\pm$ std** test error across repetitions.

Classifier \ Ratio	0.1	0.25	0.5
Independent Gaussian	0.1668 $\pm$ 0.0082	0.1637 $\pm$ 0.0069	0.1556 $\pm$ 0.0024
Multivariate Gaussian	0.3257 $\pm$ 0.0078	0.3188 $\pm$ 0.0043	0.3183 $\pm$ 0.0058
Parzen window ($h_1=0.001$)	0.3250 $\pm$ 0.0079	0.3184 $\pm$ 0.0043	0.3181 $\pm$ 0.0055

Observation: The independent Gaussian model remains the most accurate across reductions and even improves slightly at 0.5—consistent with reduced variance in parameter estimates when noisy samples are trimmed. Multivariate and Parzen are relatively flat across the tested ratios.

5. Parzen window width sweep on full TRAIN (Point 4)

Widths tested. pw = [1e-4 5e-4 1e-3 5e-3 1e-2];



Comment: very small h_1 tends to overfit (high variance), very large h_1 oversmooths (high bias). The chosen h_1 in Point 2/3 should be compared to the minimum of this sweep.

6. Changed priors + reduced TEST set (Point 5)

New priors. `apriori = [0.165 0.085 0.085 0.165 0.165 0.085 0.085 0.165];`

TEST reduction pattern. `parts_test = [1.0 0.5 0.5 1.0 1.0 0.5 0.5 1.0];`

Error rates under changed priors:

Classifier	Error
Independent Gaussian	0.1433
Multivariate Gaussian	0.3596
Parzen window ($h_1=0.001$)	0.3596

Comment: Reweighting priors to reflect class frequency shift (and reducing test samples for selected classes) benefits the independent Gaussian classifier (notably lower error), while the other two classifiers remain high in this 2-feature setting.

7. Effect of z-score normalization on 1-NN (Point 6)

1-NN error (using the same two features):

Setting	Error
Raw features	0.273
Z-score normalized	0.1552

Comment: Z-score normalization substantially improves 1-NN, as distance measures become dominated by features with larger variance if left unscaled. Normalization brings features to comparable scales and improves nearest-neighbor geometry.

8. Overall conclusions

- In the chosen 2-feature space (F3, F7), the **independent Gaussian** Bayes classifier consistently outperforms the multivariate and Parzen baselines under equal priors, under training reductions, and with changed priors.
- **Data cleansing** (removing two extreme outliers) stabilized means relative to medians—an important prerequisite for Gaussian modeling.
- **Parzen width** should be selected via the sweep (Point 4) to find a bias–variance sweet spot; the Point 2 choice ($h_1=0.001$) is sensible but may be tuned.
- **Normalization** is critical for distance-based methods: 1-NN error drops from 0.2730 to 0.1552 after z-scoring.